Project Title: Smart Modular Greenhouse System

Student Name: Hasan Zahra

Student ID: [6826]

Group Number: [C3]

1) Individual Contribution Summary

Primary Role & Core Responsibilities

My primary role in this project was Lead Hardware Integrator, Firmware Developer, and Calibration Specialist. I was responsible for the complete physical assembly of the greenhouse system, developing the entire integrated software codebase, and performing rigorous calibration and system testing of all integrated sensors and actuators.

Key Technical Contributions

1) Complete System Assembly:

-Physically assembled the modular greenhouse chassis and mounted all components.

-Wired all electrical connections between the microcontroller, sensors, and high-power actuators (safely integrating relays for the 220V/high-current water pump and cooling fan)

2) Software Design & Architecture:

-Wrote and structured the complete modular code across greenhouse. No, config.h, and greenhouse_sensors.h.

-Programmed the core automation loops, sensor-reading cycles, and actuator thresholds.

3) Sensor Calibration & Unit Testing:

-Soil Moisture: Calibrated raw analog ranges (1002 to 360) to map 0% to 100% soil saturation.

-DHT11 Temp/Humidity: Integrated the sensor and programmed thermal response safety limits (e.g., fan cooling trigger at 34 degrees Celsius).

-LDR (Light): Calibrated ambient light thresholds (350 limit) to automatically trigger the digital grow lamp in low-light conditions.

-Water Flow & Pump: Calibrated the flow sensor using a 5.59 pulse factor to monitor exact water delivery during active pump cycles.

4)Corrosion Mitigation Power Gating:

-Programmed a custom power-saving cycle for the soil moisture sensor, using a digital pin (Pin 6) to power the sensor only during the brief 200ms active reading window to prevent electrolysis and probe degradation.

Verification & Validation

I executed unit-test protocols for each isolated sensor over designated testing intervals to ensure data accuracy before final system integration. The successful graphs and logs generated from these tests validate that the physical assembly and software are fully operational.

2) Personal Reflection Student

Challenges Faced & Practical Solutions

As the person responsible for both the physical assembly and the programming, one of the most critical challenges I faced was managing sensor degradation and electrical efficiency. Soil moisture sensors are notorious for corroding rapidly due to electrolysis when constantly powered with 5V in damp soil.

To solve this hardware limitation, I implemented a software-controlled "power gating" technique. I wired the sensor's Vcc pin to a digital pin (Pin 6) instead of constant power. In the code, I programmed the microcontroller to pull Pin 6 high, wait a brief 200ms for the voltage to stabilize, take the analog reading, and immediately pull the pin low again. This simple workaround completely protects the physical probes from rapid corrosion and ensures the long-term reliability of our greenhouse system.

2. Key Lessons Learned

1)The Value of Assembly Planning:

I learned that clean wiring and secure sensor mounts are just as important as clean code. Organizing the power distribution safely for both low-power sensors and high-current relays (like the pump and fan) was critical to preventing electrical noise and sudden system resets.

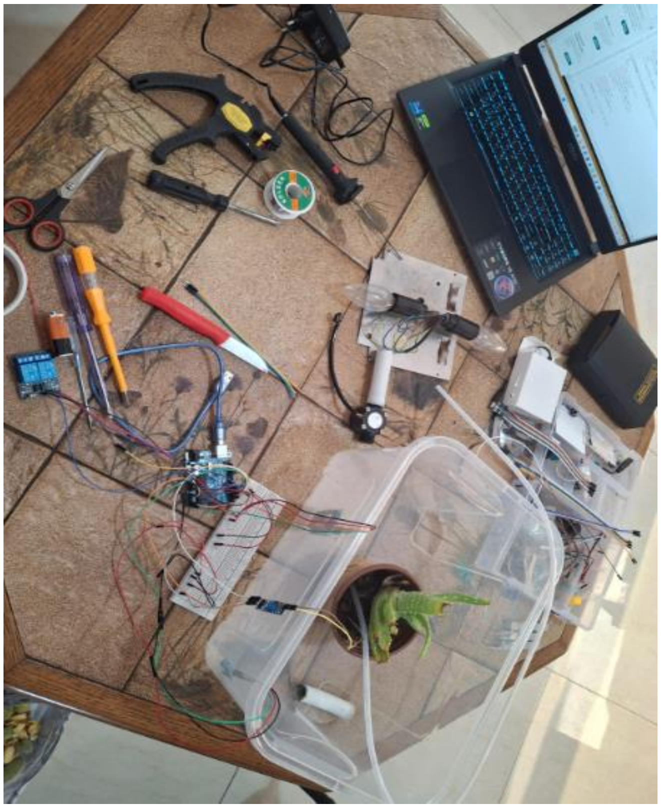
2)Modular Unit Testing:

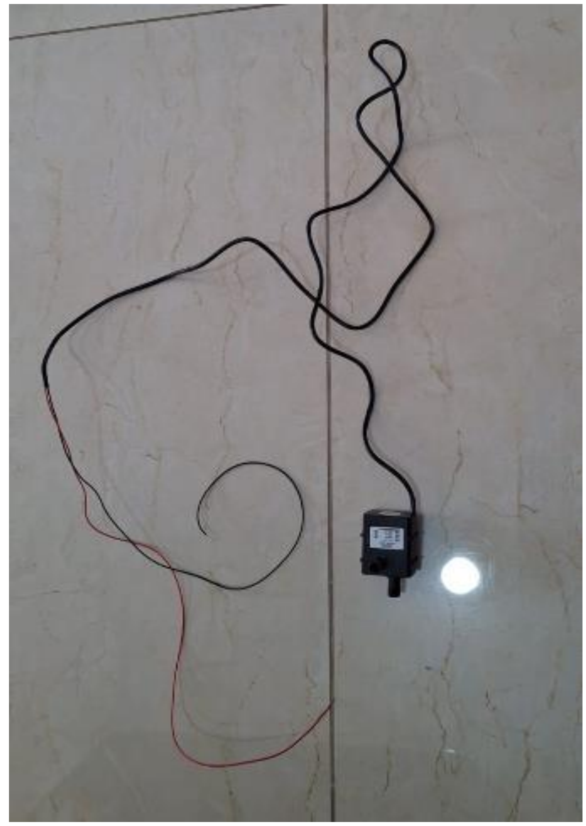
Trying to debug a fully assembled greenhouse all at once is nearly impossible. I learned that writing standalone test codes for each sensor, recording the data, and graphing the outputs in Excel over 60-second intervals is the most effective way to guarantee every physical connection and code line is solid before deployment.

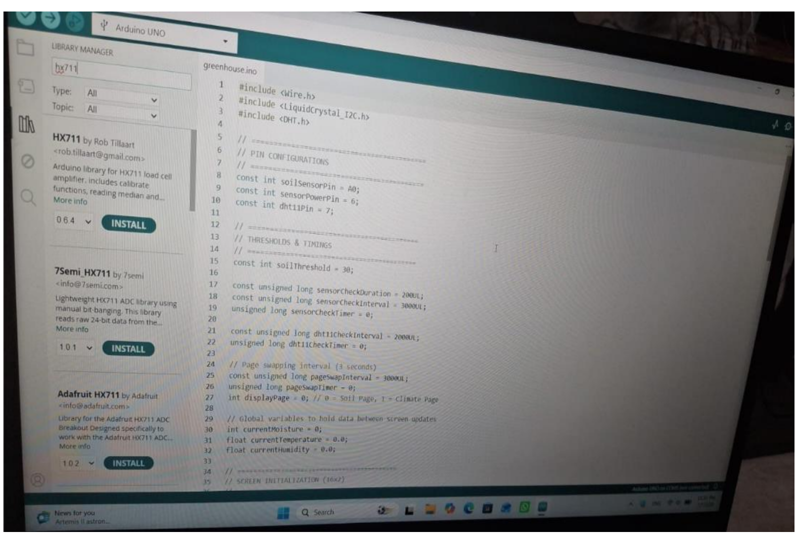
3)Project Management & Responsibility:

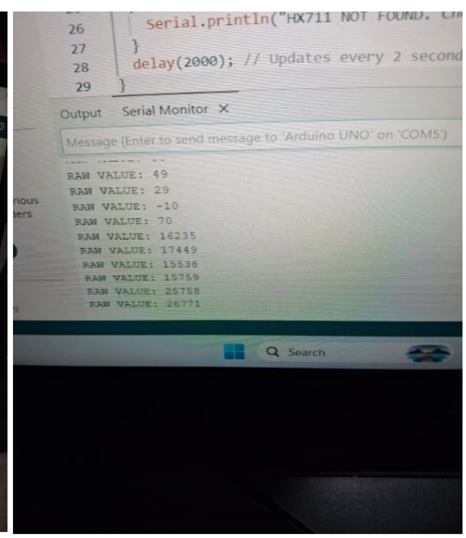
Taking ownership of the physical assembly, coding, and testing gave me a deep, systems-level understanding of the entire project. It taught me how critical it is to document calibration steps so that the rest of the team can easily interpret the system's physical data.

3) Evidence of work performed






Arduino UNO
LIBRARY MANAGER
hx711
Type: All
Topic: All
HX711 by Rob Tillaart
Arduino library for HX711 load cell amplifier. Includes calibrate functions, reading median and...
More info
INSTALL
7Semi_HX711 by 7semi
Lightweight HX711 ADC library using manual bit banging. This library reads raw 24-bit data from the...
More info
INSTALL
Adafruit HX711 by Adafruit
INSTALL
greenhouse.ino
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>
// PIN CONFIGURATIONS
const int soilSensorPin = A0;
const int sensorPowerPin = 6;
const int dht11Pin = 7;
// THRESHOLDS & TIMINGS
const int soilThreshold = 30;
const unsigned long sensorCheckDuration = 200UL;
const unsigned long sensorCheckInterval = 3000UL;
unsigned long sensorCheckTimer = 0;
const unsigned long dht11CheckInterval = 2000UL;
unsigned long dht11CheckTimer = 0;
// Page swapping interval (3 seconds)
const unsigned long pageSwapInterval = 3000UL;
unsigned long pageSwapTimer = 0;
int displayPage = 0; // 0 = Soil Page, 1 = Climate Page
// Global variables to hold data between screen updates
int currentMoisture = 0;
float currentTemperature = 0.0;
float currentHumidity = 0.0;
// SCREEN INITIALIZATION (16x2)
Search


Serial.println("HX711 NOT FOUND.
}
delay(2000); // Updates every 2 second
}
Output
Serial Monitor
Message (Enter to send message to 'Arduino UNO' on 'COM5')
RAW VALUE: 49
RAW VALUE: 29
RAW VALUE: -10
RAW VALUE: 70
RAW VALUE: 16235
RAW VALUE: 17449
RAW VALUE: 15538
RAW VALUE: 15759
RAW VALUE: 25758
RAW VALUE: 26771
Search